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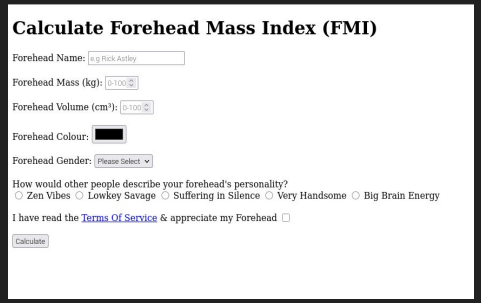
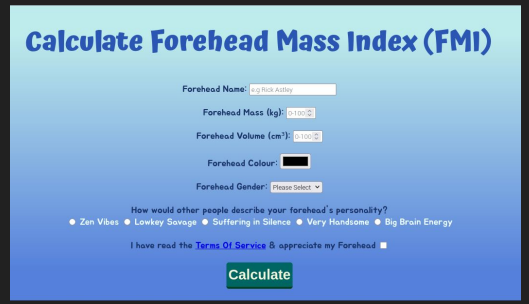
Forehead Mass Index (FMI)

```
29      switch (personality) {
30          case "Zen-Vibes":
31              personalityValue = -10;
32              break;
33          case "Lowkey-Savage":
34              personalityValue = -5;
35              break;
36          case "Very-Handsome":
37              personalityValue = 5;
38              break;
39          case "Big-Brain-Energy":
40              personalityValue = 10;
41              break;
42          default:
43              break;
44      }
45      // Modifies personalityValue based on input
```





What?

- A website that calculates someone's Forehead Mass Index (FMI)
- FMI is a fictional metric inspired by Body Mass Index (BMI)

How?

- Developed with Web Technologies:
  - **HTML**
  - **CSS**
  - **JavaScript**
- **Git** for Version Control & **GitHub** for hosting

Results

- Explore the website here: https://3omargaad.github.io/FMI/
- 5 types of FMI are available, can you find them all?

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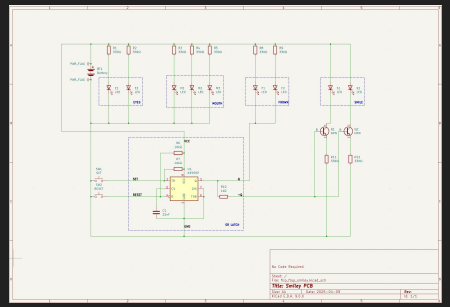
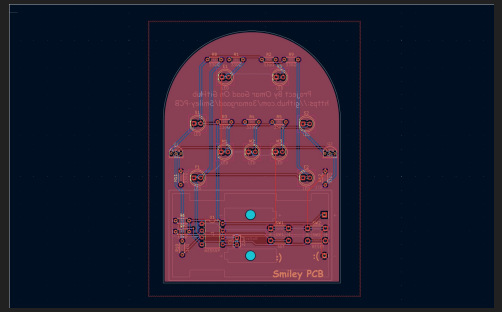
Smiley PCB





What?

- A circuit board that starts of frowning, but smiles with the push of a button

How?

- Prototyped on a **Breadboard**
- **Schematic & PCB Design** developed using **KiCad** Software
- **Ordered** online from a PCB Manufacturing company
- **Soldered** THT Components by hand

Results

- Functional Circuit using only 2 AA Batteries
- Parallel Design ensures if one LED breaks, the rest of the circuit works as expected

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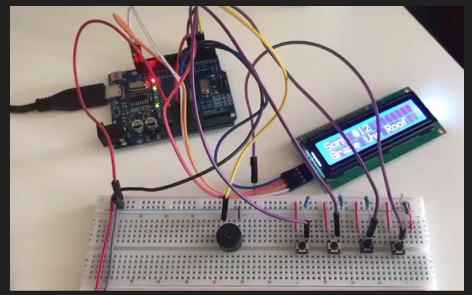
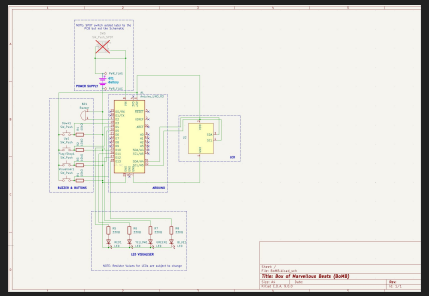
Box of Music Beats (BoMB)





What?

- An electronic music box that plays retro tones such as the Tetris and Mario theme.
- *Disclaimer: This product is not intended to explode*

How?

- Device uses **Arduino Microcontroller**
- **PlatformIO** for software debugging
- Worked with an **I2C LCD Module** for navigation
- **3D Printed** case, modelled using **FreeCAD** software
- **Collaborated** with a friend

Results

- Up to 15 songs stored on the device
- Currently works without LCD, progress being made to implement it

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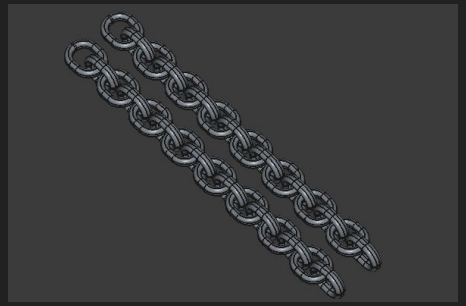
DIY Watch Chain



What?

- I repeatedly broken my watch strap as a result of fiddling with it
- Instead of re-fixing it, I decided to create a custom watch chain

How?

- Used **FreeCAD** to **3D model chain**
- Printed the main chain using my school's **3D Printer**
- Ordered 'S Link' parts from 3D Printing Manufacturer

Results

- A satisfying, fiddle-proof watch
- Concept reduces E-waste
- 'S Link's made out of PETG instead of PLA for durability

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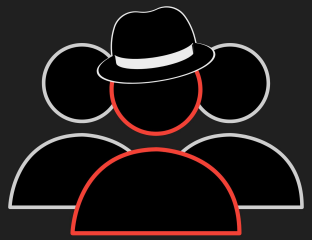
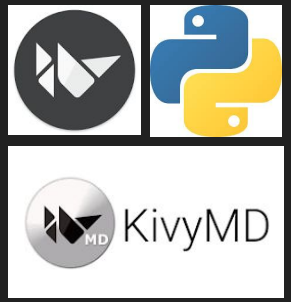
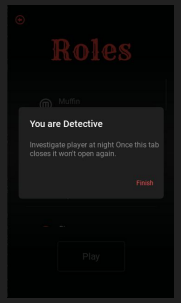
Mafia







What?

- A mobile and desktop cross-platform application for the social deduction game
- Your device acts as the “host”, allowing everyone to play.

How?

- Programmed in **Python** using the **kivymd** library for UI
- Project **documentation** to record analysis, design and testing stages
- Created an **Android Developer** account to publish mobile app

Results

- The app GUI has been developed
- Functionality in progress...